

Tianxin Hu

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Educational Background

Nanyang Technological University (Master of Science) - School: Electrical and Electronic Engineering - Major: Computer Control & Automation - GPA: 95.00/100	Aug 2024 – Jan 2026
Wuhan University of Science and Technology (Bachelor of Engineering) - School: Automobile and Traffic Engineering - Major: Vehicle Engineering (Industrial Planning) - GPA: 88.30/100 (Class Ranking: 3/32 Department Ranking: 8/157)	Sept 2020 – Jun 2024

Research & Academic Experience

Chery Robotics — Humanoid Robot Locomotion Development Supervisor: Professor Lihua Xie Position: Research Assistant Project: Industry-oriented humanoid robotics project targeting language-guided embodied task execution, such as object fetching in real-world environments Responsibilities: Responsible for the locomotion module of Chery's humanoid robot, with a focus on reinforcement learning-based humanoid locomotion training and deployment	Jun 2026 – Present
AIBOT Technology Pte. Ltd. — Autonomous Navigation System Development Supervisor: Professor Lihua Xie Position: Navigation Module Lead Responsibilities: Developed the full-stack navigation module for the company's autonomous mobile robot, responsible for cross-floor navigation, elevator interface integration, and dynamic obstacle avoidance within residential and commercial environments ✓ Achievements: Achieved fully autonomous navigation and delivery within multi-story residential complexes	Jul 2025 – Jun 2026
Research Student — NTU Internet of Things Lab & Center for Advanced Robotics Technology and Innovation (CARTIN) Supervisors: Professor Lihua Xie Research Focus: Developed optimization-based motion planning methods for multi-axle autonomous vehicles to minimize swept volume during turning maneuvers and reduce tire wear under dynamic constraints ✓ Achievements: Published four papers on planning and control [J1-J5] and received a national award [A1]	Aug 2024 – Mar 2026
Xin Chi Technology — Research and Development of Autonomous Driving System for Multi-Axle Heavy-Duty Transport Vehicles Supervisor: Senior Engineer Xiangrong You Position: Technical Director & Corporate Legal Representative Responsibilities: Led R&D of autonomous driving systems for heavy-duty vehicles, covering mathematical modeling, ROS architecture, microcontroller development, and both simulation and prototype validation ✓ Achievements: Contributed to multiple patents [P1-P4] and received several national awards [A2-A6]	Jun 2022 – Jun 2024
Research and Development of Autonomous Driving System for Formula Racing Supervisors: Associate Professor Chao Deng, Dr. Junyi Zou Position: Leader of the Unmanned Group Responsibilities: Led the development of the autonomous driving system for Formula Student Autonomous China (FSAC), covering visual detection, mapping, planning, and tracking ✓ Achievements: Received the FSAC Third Prize and Best Rookie Award [A7], and published a paper on it [J6]	Feb 2021 – Jun 2024

Publications

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- [J1] **Multi-Agent Off-World Exploration for Sparse Evidence Discovery via Gaussian Belief Mapping and Dual-Domain Coverage** Jun 2026
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)
Zhuoran Qiao, **Tianxin Hu**, Thien-Minh Nguyen, Shenghai Yuan
DOI: [10.48550/arXiv.2603.07650](https://doi.org/10.48550/arXiv.2603.07650)
- [J2] **Energy-Constrained Navigation for Planetary Rovers With Singular Internal Power Source** Apr 2026
IEEE International Conference on Control & Automation (ICCA 2026 ORAL)
Tianxin Hu, Weixiang Guo, Rui Qian, Jiaye Jin, Haoran Zhao, Shenghai Yuan, Lihua Xie
- [J3] **Swept Volume-Aware Trajectory Planning and MPC Tracking for Multi-Axle Swerve-Drive AMRs** May 2025
IEEE International Conference on Robotics and Automation (ICRA 2025)
Tianxin Hu, Shenghai Yuan, Ruofei Bai, Xinhang Xu, Yuwen Liao, Fen Liu, Lihua Xie
DOI: [10.1109/ICRA55743.2025.11127914](https://doi.org/10.1109/ICRA55743.2025.11127914)
- [J4] **Tire Wear-Aware Trajectory Tracking Control for Multi-Axle Swerve-Drive Autonomous Mobile Robots** Jun 2025
Journal of Automation and Intelligence (JAI)
Tianxin Hu, Xinhang Xu, Thien-Minh Nguyen, Fen Liu, Shenghai Yuan, Lihua Xie
DOI: [10.1016/j.jai.2025.05.003](https://doi.org/10.1016/j.jai.2025.05.003)
- [J5] **TGSFormer: Scalable Temporal Gaussian Splatting for Embodied Semantic Scene Completion** Mar 2026
IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2026)
Rui Qian, Haozhi Cao, Tianchen Deng, **Tianxin Hu**, Weixing Guo, Shenghai Yuan, Lihua Xie
DOI: [10.48550/arXiv.2512.00300](https://doi.org/10.48550/arXiv.2512.00300)
- [J6] **Improvement of YOLOv5 Algorithm for Formula Racing Based on TensorRT under ROS Platform** May 2023
Agricultural Equipment & Vehicle Engineering
Tianxin Hu, Chao Deng, Junjie Ma, Wang Liu
DOI: [10.3969/j.issn.1673-3142.2023.05.004](https://doi.org/10.3969/j.issn.1673-3142.2023.05.004)

Patents

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- [P1] **(Granted) Multi-axle all-wheel steering vehicle vector-division correction method and system** Sep 2025
Tianxian Hu, Xiangrong You, Dongmei Wang, Xinhe Long, Xinyu Wang, Shuang Liu, Yutong Fu, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Patent No.: [CN116534005B](#)
- [P2] **(Granted) A system and method for automatically correcting the deviation of a beam transport vehicle in a tunnel** Jul 2023
Xiangrong You, Enze Jin, Dongmei Wang, Yuran Huang, **Tianxian Hu**, Shuai Zhu, Beijia Zhang
Patent No.: [CN114633801B](#)
- [P3] **(Pending) Energy-saving system and method for beam transport vehicle based on traveling pressure control** Feb 2024
Wei Li, Xiangrong You, Dongmei Wang, Xinhe Wang, Peng Cheng, Yuming Liu, Shuang Hu, **Tianxian Hu**, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Application No.: [CN117508238A](#)
- [P4] **(Pending) Method, system and equipment for planning obstacle avoidance path of multi-shaft all-wheel steering beam transporting vehicle** Dec 2023
Xiangrong You, Yucheng Yin, Liwei Wang, Dongmei Wang, **Tianxian Hu**, Shuang Liu, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Application No.: [CN117168481A](#)

Awards

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- [A1] **(International Level) Best Poster Award — ICRA 2026 Workshop** Jun 2026
Robots Meet Prior Maps: Leveraging Geometric and Semantic Priors for Real-World Robot Autonomy, Vienna, Austria

[A2]	(International Level) Bronze Award of the 9th China International College Students Innovation Competition <i>Organizing Committee of China College Students “Internet+” Competition</i>	Apr 2024
[A3]	(National Level) The First Prize of the 18th “Challenge Cup” National Undergraduate Curricular Academic Science and Technology Works <i>China Association for Science and Technology, Ministry of Education of the PRC, Chinese Academy of Social Sciences</i>	Oct 2023
[A4]	(International Level) Bronze Award of the 8th China College Students “Internet+” Innovation and Entrepreneurship Competition <i>Organizing Committee of China College Students “Internet+” Competition</i>	Apr 2023
[A5]	(National Level) Second Prize of the 15th China College Student Computer Design Competition <i>Organizing Committee of Chinese College Student Computer Design Competition</i>	Jul 2022
[A6]	(National Level) Second Prize of the 15th Energy Saving and Emission Reduction Competition <i>China College Student Energy Conservation and Emission Reduction Social Practice and Technology Competition Committee</i>	Aug 2022
[A7]	(National Level) Third Prize and Best Rookie Award of Formula Student Competition <i>China Society of Automotive Engineers, Organizing Committee of FSAC</i>	Oct 2022

Technical Expertise

Programming: C, C++, Python, XML (URDF/Launch)

Robotics & Simulation: ROS/ROS2, Isaac Sim/Lab, Gazebo Classic & GZ Sim, RViz, URDF/SDF modeling

Embedded Systems: NVIDIA Jetson Series, STM32, ESP8266, 8051; FreeRTOS, LVGL, U8G2

Hardware Design: Altium Designer, Proteus; multilayer PCB design and soldering

Modeling, Simulation & CAD: MATLAB, COMSOL, SolidWorks, CATIA, Blender, AutoCAD